Retrospective Sprint 1 of Group UG13

Block Model Compression Algorithm

Team Members:

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3. a1824243 Yuyao Chen
4. a1837254 Yang Shi
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Snapshots

I attended the sprint review/planning meeting on 21/Aug with the tutor.

Snapshots are provided in the appendix at the end of the retrospective.

What went well in the sprint?

The sprint went on with positive overtones due to good teamwork and good communication principles including sharing ideas and understanding the strengths and visions of other team members. In such an ambience, members supported each other and felt their contribution was important.

Moreover, the team was best able to arrange meeting preparation according to individual strengths. This allowed us to save time in preparing and improving the quality of our meeting by ensuring we had the information covered and ready to go on while streamlining our discussions.

The other key to success in this sprint was the active participation of every member in the discussions. The members were encouraged to actively participate in sharing both technical (such as algorithm's performance and efficiency) and non-technical (namely client requirements) challenges, questions, and ideas. This not only gave rise to more strong and vigorous solutions but also a sense of collective ownership to find these solutions.

In all, effective teamwork, efficiency at meetings, and active participation made this sprint very successful. All these factors also gave a good initiation concerning all the upcoming sprints, showing the team can work together towards achieving a common and meaningful objective.

What could be improved?

Firstly, I believe the organization of the GitHub Project can be improved. More explanation is provided in the next heading (What will the group commit to improve in the next sprint?).

Secondly, at times, discussions became overly focused on specific issues that were interdependent with others, causing the discussion to be all over the place and diverting attention to details prematurely. I believe this issue could have been mitigated by clearly defining the topic before each discussion. For instance, when we discussed the compression techniques to be used, our focus shifted to the specifics of GPU vs. CPU processing, such as memory management and efficiency, which were interdependent with other aspects like the complexity of the algorithm and the input and output of data. To avoid this, we should first focus on identifying the compression techniques to explore, before diving into the technical specifics. This structured approach would have ensured that each discussion remained focused and productive.

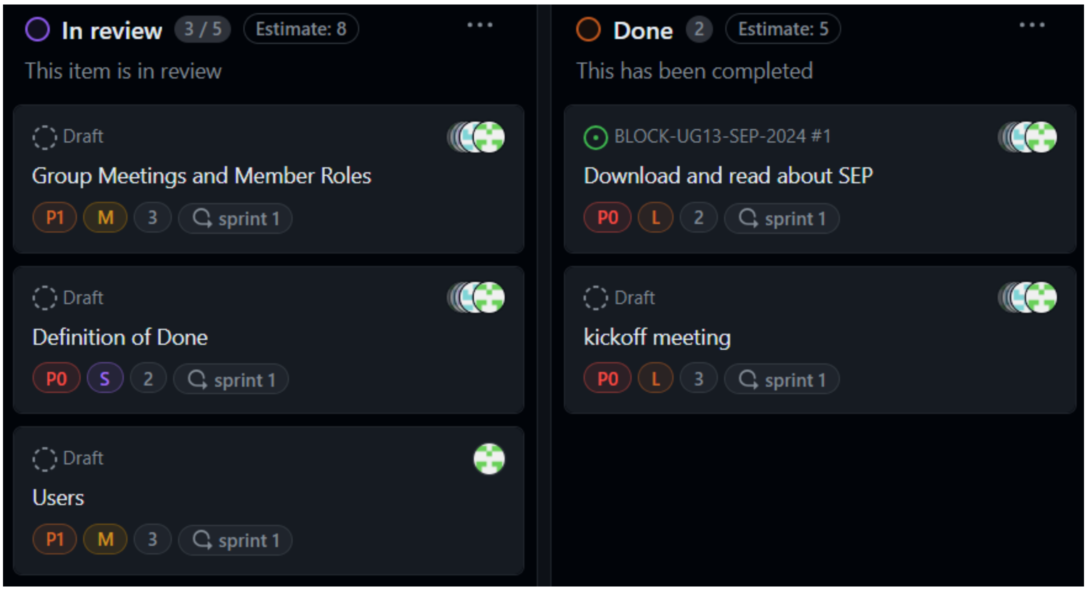
What will the group commit to improve in the next sprint?

I suggest our group improve the organization and documentation of the GitHub Project in the next sprint.

Our team maintained the product backlog, sprint backlog, and a task board initially with an online spreadsheet. However, we have become more familiar with GitHub Projects and thought it would be a better way to present our progress to the proxy client. Although we have already moved all tickets and their information to GitHub Projects, I would recommend several enhancements in task board organization.

Firstly, I suggest renaming the status “Backlog" to "Product Backlog” and "Ready" to "Sprint Backlog” to eliminate vagueness and align with Agile terminology. Secondly, descriptions written on the tickets should be more descriptive, and acceptance criteria can be added for a more technical definition of done, as the proxy client suggested. Finally, we'll import the shared documents from the communication channel (group chat) into GitHub Wiki so it will be more easily assessable, as all the reference materials will be collected in one place.

My progress this sprint



The main goals for our team in the first sprint are: gathering requirements, creating clear and concise API documentation, and making the program read input in the given format correctly. To effectively achieve these goals, the goals are subdivided into smaller tasks, which are then assigned to team members.

The screenshot above shows the tasks assigned to me in this sprint, which include scheduling meetings, contributing to the definition of done, identifying target users and their user stories, reading the Scrum Event Planning guide, and summarizing the kickoff meeting.

Firstly, scheduling meetings involves consulting the time availability of team members, reminding the objectives of the meeting to team members, and confirming the time and place of the meeting. Secondly, I contribute to the definition of done by reminding the team that we should ensure the design and API documentation are updated after the change in code passes all the tests so that they stay relevant. Thirdly, I brainstormed and documented the possible users for the program and their user stories, for example:

- As a client, I want the program to be able to read the input in the specified format and produce a compression output without loss correctly and efficiently.
- As a developer, I want to create an optimized compression algorithm that can handle large block models, ensuring my application performs well even with extensive data.

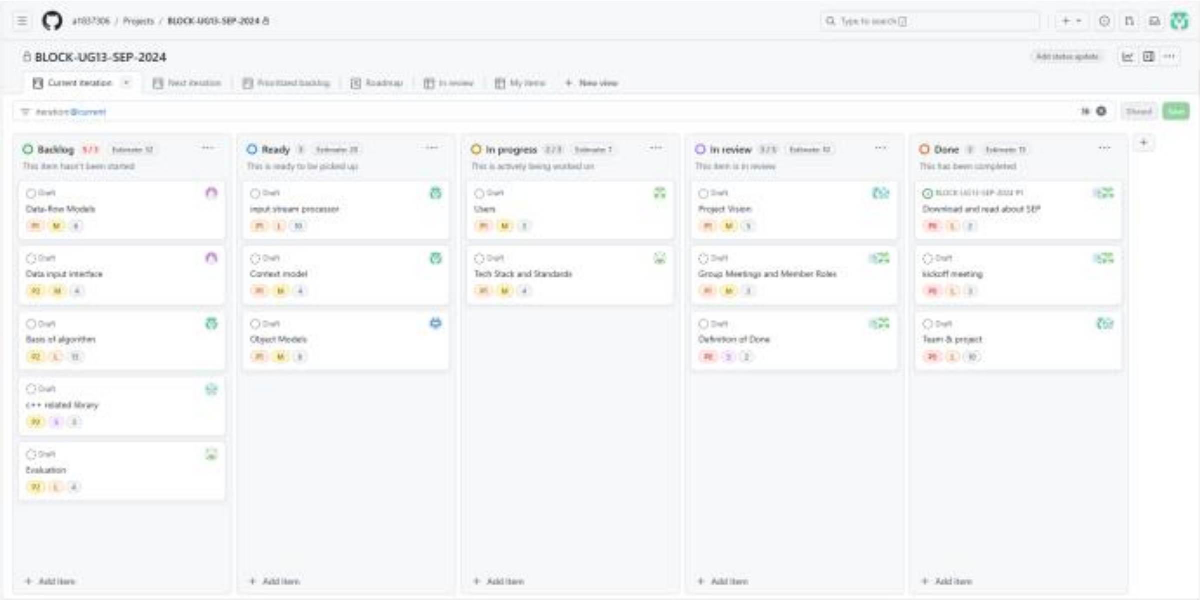
There are other users I identified such as 3D modelers, geologists, and database administrators. However, they can be classified under the client umbrella for simplicity since their user stories are mostly similar. The client user story above can be subdivided into smaller user stories to represent each user requirement and to incorporate into the weekly snapshot.

Fourthly, I reviewed the Scrum Event Planning guide to be on the same page as the team in terms of the agenda and activities for the Scrum ceremonies. This makes the development process smoother. Lastly, I summarized the kickoff meeting and shared the summary with the team so that those who couldn’t attend the meeting could pick up the information quickly, while those who attended could have a fast and clear revision of what had been discussed in the meeting. The summary includes how to work with GitHub Project, what is the required information for each ticket on the task board, and clarifications on project requirements.

Appendix

Week 5 Snapshot

Product Backlog and Task Board:



Sprint Backlog and User Stories:

Sprint Overview

For the current sprint, we have selected the following user stories to focus on, catering to both developers and clients:

1. As a developer, I want to have a clear and concise API documentation so that I can quickly understand and implement the block model compression algorithm.

- Task 1.1: Create detailed API documentation for the compression algorithm.

- Task 1.2: Ensure documentation covers all public methods and parameters.

- Task 1.3: Provide code examples for common use cases.

2. As a client, I want to be able to easily import my block models into the program so that I can begin the compression process without any hassle.

- Task 2.1: Design a user-friendly import feature for block models.

- Task 2.2: Ensure compatibility with various block model formats.

- Task 2.3: Implement error handling for import issues.

These user stories and their related tasks will guide our development efforts for the current sprint, ensuring that we address the needs of both developers and clients in creating an effective block model compression program.

ID	Title	Description	Priority	Estimated Effort	Responsible
SB-001	Scrum tool	Register, try, compare and choose Scrum management tools	2	2	
SB-002	API documentation	Create detailed API documentation for the compression algorithm.	1	3	
SB-003	Methods & parameters	Ensure documentation covers all public methods and parameters.	1	4	
SB-004	Code examples	Provide code examples for common use cases.	2	5	
SB-005	Import feature	Design a user-friendly import feature for block models.	1	3	
SB-006	Compatibility	Ensure compatibility with various block model formats.	2	4	
SB-007	Error handling	Implement error handling for import issues.	2	5	

Definition of Done

1. Code complete:

All planned functionality has been implemented, and the code conforms to the team's coding specifications.

2. Code Review passed:

The code has been peer reviewed at least once and all issues raised have been addressed.

3. Compile pass:

Compile through, there are no errors reported, and there are no warnings that affect the operation.

4. Tests passed:

All relevant automated tests (e.g., unit tests, integration tests) pass, and test coverage meets the standards agreed upon by the team.

All relevant manual tests (e.g., functional tests, acceptance tests) passed and no new defects were found.

5. Performance tests passed:

If applicable, all relevant performance tests have been passed and the performance metrics meet the standards agreed upon by the team.

6. Documentation updates:

All relevant technical documentation (such as design documentation, API documentation) has been updated and is consistent with the code.

7. Code merged into the main branch:

The code has been merged into the main branch and no new conflicts or problems have been introduced.

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PB-001	Download and read about SEP	Download and study the Scrum Event Planning guide	1	2	
PB-	kickoff meeting	Organize and conduct the project kick-off meeting	1	3	
PB-	Team & project	Assemble the Scrum team and select a suitable project	1	4	
PB-	Project Vision	Define the overall goal and desired outcome of the project	2	5	
PB-	Users	Identify and document the target users for the project	2	3	
PB-006	Context model	Create a context model to understand the environment and constraints of the project	3	4	
PB-007	Data-flow Models	Develop data flow models to visualize the flow of data within the system	3	5	
PB-008	Object Models	Design object models to represent the structure and behavior of the system	3	6	
PB-009	Tech Stack and Standards	Choose the technology stack and establish coding standards	2	4	
PB-010	Group Meetings and Member Roles	Schedule regular meetings and define roles and responsibilities for team members	2	3	
PB-	Data input interface	Plan the design and implementation of the data input interfaces	2	4	
PB-	Basis of algorithm	Establish the basis for the algorithms used in the project	4	5	
PB-	c++ related library	Research and select appropriate libraries for C++ development	3	3	
PB-014	Evaluation	Evaluate the progress and results of the project against the defined goals	5	4	

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PB-002	kickoff meeting	Organize and conduct the project kick-off meeting	1	3			Done	
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PB-004	Project Vision	Define the overall goal and desired outcome of the project	2	5			Done	
PB-005	Users	Identify and document the target users for the project	2	3		In Progress		
PB-006	Context model	Create a context model to understand the environment and constraints of the project	3	4		In Progress		
PB-007	Data-flow Models	Develop data flow models to visualize the flow of data within the system	3	5		In Progress		
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PB-010	Group Meetings and Member Roles	Schedule regular meetings and define roles and responsibilities for team members	2	3			Done	
PB-011	Data input interface	Plan the design and implementation of the data input interfaces	2	4	To Do			
PB-012	Basis of algorithm	Establish the basis for the algorithms used in the project	4	5	To Do			
PB-013	c++ related library	Research and select appropriate libraries for C++ development	3	3	To Do			
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Summary of Changes:

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Week 6 Snapshot

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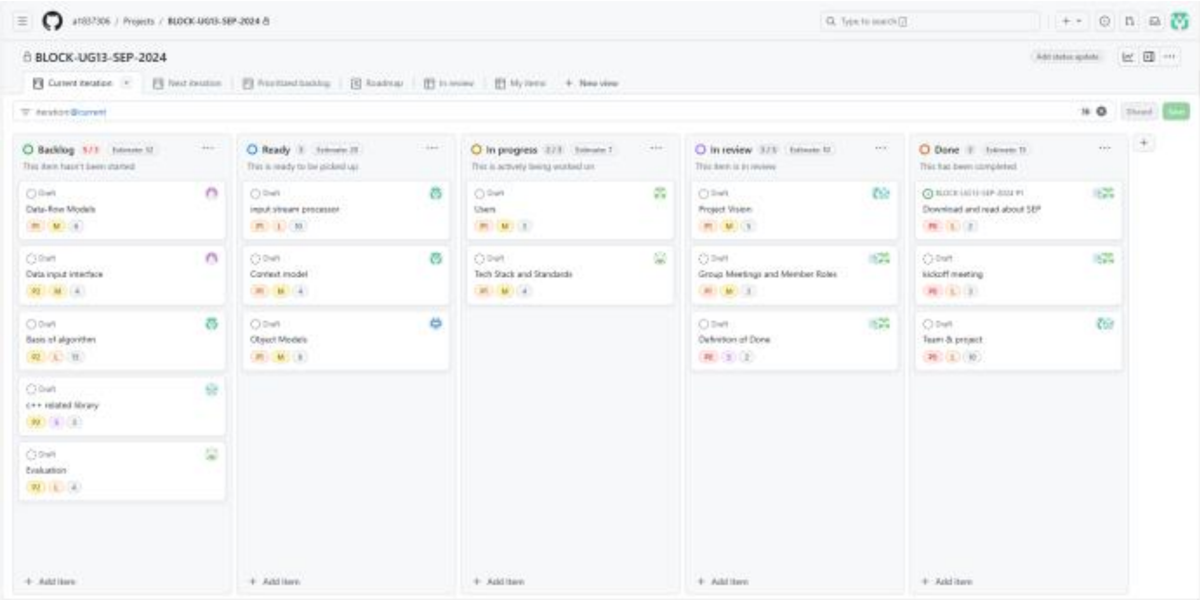

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